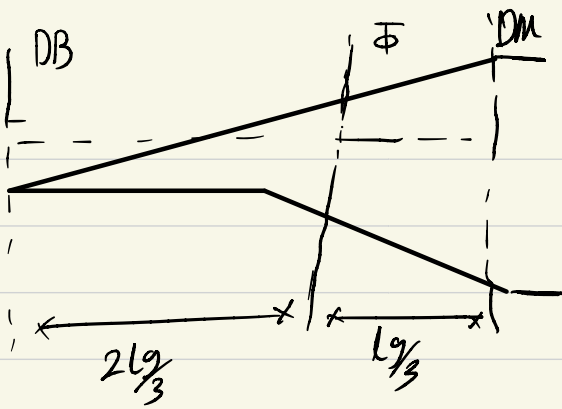
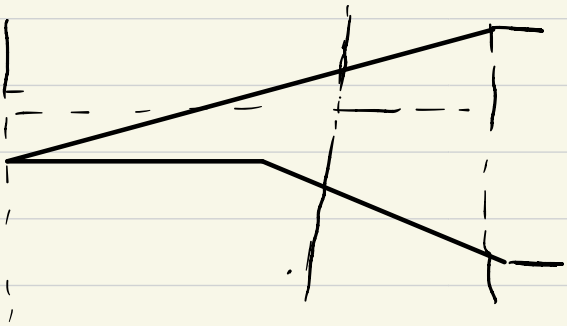
Fine

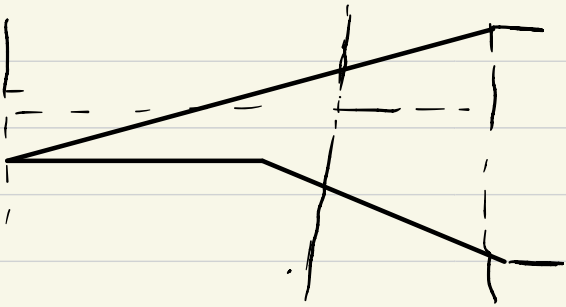
super elevation
without Transion curve



super elevation
with Transion curve
but without ramp

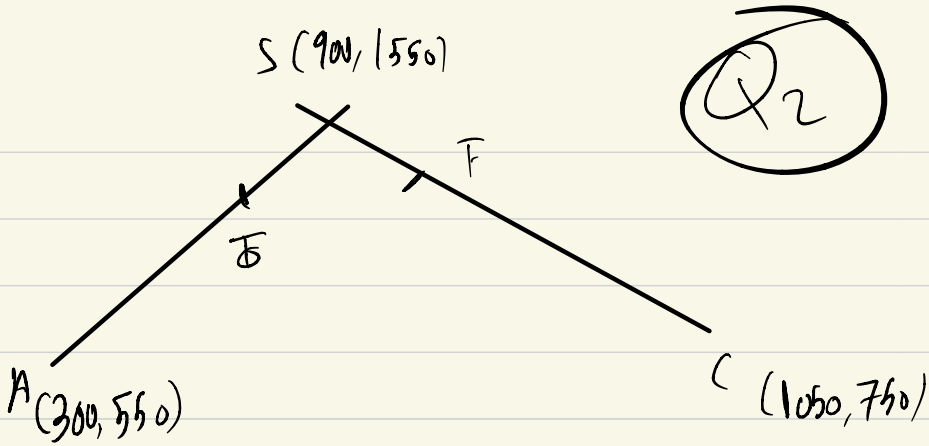


super elevation
with Transion curve
With ramp



*Write the important of the intersection of a road (write two sentences) ?

*when does the inflection Poin(DN) and bisector point (B)?



* Find (|AS|, |SC|) and the angle (Δ) ?

* Find Horizontal curve elements with minimum possible Radius, half of the centrefugal force is used

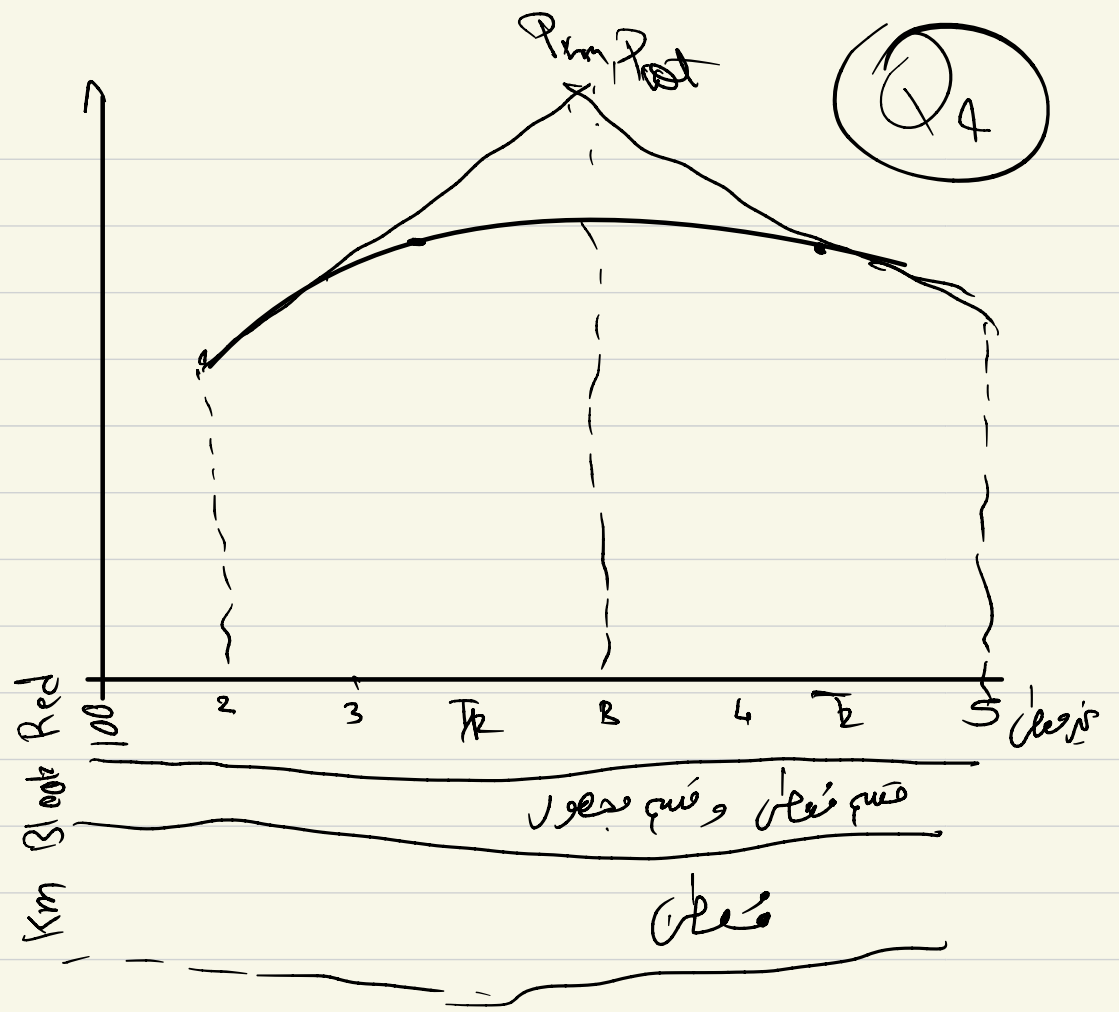
* Find the minimum clothiod paramete, with Transion curve length, with ramp ?

* Draw the F corss-section, knowing that the Red elevation (393) and black elevation 370 ?

Q3

* يوجد شبيهه بالأ يقولوما

$T_n = T_0(1+r)^n$



① Find (T_{1km}, P_{km}, T_{2km}) and their elevations?
② Find the unknowns above (Red elevation)
③ Find (ΔN_{km}) + (ΔN_{ele}) ?